Minkyu Kim

minkyu.kim@snu.ac.kr | +82 10 2344 7000 | minkyu-kim-research.github.io

Education

Seoul National University — Seoul, South Korea
Ph.D. in Intelligence and Information (in progress) — Sept 2026 – Present
Advised by Prof. Nojun Kwak

M.S. in Engineering, Department of Intelligence and Information — Sept 2024 – Aug 2026
- GPA 4.11/4.30. Advised by Prof. Nojun Kwak

B.S. in Mechanical Engineering & Interdisciplinary Major in Artificial Intelligence — Mar 2019 – Aug 2024
- GPA 3.86/4.3 (Major GPA 3.90, Interdisciplinary Major GPA 3.94)
- 1.5 years of absence due to obligatory military service (2021 - 2022)

Incheon Academy of Science and Arts — Incheon, South Korea
- High school for talented students in science and mathematics — Mar 2016 – Feb 2019

Research Experience

Recursive Harness Distillation for Robot Manipulation — Aug 2026 – Present
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak
- Contributed to method design discussions on transferring robot intervention experience across agents through a reusable playbook refined with execution feedback.
- Supported physical-robot experiments for the shared system, which improved manipulation success from 37.3% to 64.0% without updating model parameters.

Adaptive VLA Action-Chunk Execution (PAV) — May 2026 – Present
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak
- Developed a policy-aligned predictive verifier that scores candidate action prefixes in the VLA's task-conditioned representation space before execution.
- Learned prefix validity from terminal outcomes using survival supervision and a localization prior. Evaluated success–query trade-offs with π_0 and $\pi_{0.5}$ on LIBERO, RoboTwin 2.0, and real robots.

Completion Aware Guidance for World Action Models — Oct 2025 – Sept 2026
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak
- Co-authored CAG, a training-free sampling method that strengthens relevant instruction conditioning to address task-incomplete imagination in world action models.
- The method improved success from 64.4% to 70.0% on a RoboTwin 2.0 subset and from 69% to 75% in zero-shot simulation with Fast-WAM and DreamZero.

High-Throughput Diffusion Generation (ManiFusion) — Feb 2025 – May 2026
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak and Prof. Jangho Kim (Kookmin University)
- Developed fusion/unfusion modules that share backbone computation across multiple samples, with a geometric analysis of recoverable sample information.
- Introduced Trajectory-Fusion Scheduling to vary the fusion factor across denoising stages. Evaluated U-Net, DiT, and autoencoder variants across diffusion and flow-matching models.

Drone-Based Marine Safety Management — Sept – Dec 2024
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak
- Developed a YOLO-based system for detecting threats near humans, using post-training quantization for on-device deployment.

Dermatological Visual Question Answering — Jan – June 2024
MIPAL, Seoul National University. Advised by Prof. Nojun Kwak
- Fine-tuned MedVInT-Decoder, LLaVA, and LLaVA-Med for medical answer generation, placing second among teams in the MEDIQA-M3G English track.
- Evaluated synthetic query–response augmentation and its limitations. Post-challenge LLaVA achieved a BLEU score of 6.98.

Generative Models for Journal-Bearing Fault Detection — Jan – Aug 2024
Hyperautomation Artificial Intelligence Lab, Seoul National University. Advised by Prof. Byeng Dong Youn

- Synthesized abnormal vibration data to address class imbalance. Received the Outstanding BS Thesis Presentation Award.

Slope Control for Electric Walking Aids — Mar – Dec 2017
Motion Control Lab, Yonsei University. Advised by Prof. Yoon Su Baek

- Used FSR foot sensors to infer walking intent and speed for slope-dependent speed compensation.

Bio-Inspired Retractable Dome Design — Aug 2016 – Mar 2017
Incheon Academy of Science and Arts.

- Designed a dome inspired by Mugunghwa petal arrangements. Selected for Korea's Intel ISEF 2017 delegation.

Publications

Minkyu Kim, Baekseung Kim, Junhoo Lee, Jangho Kim, Nojun Kwak, ManiFusion: Unlocking the High-Throughput Generation via Superposition in Manifold Space, NeurIPS, 2026 (Accepted).

Minkyu Kim, Junhoo Lee, Seungyeon Kim, Suhyun Jeon, Baekseung Kim, Donghoon Han, Nojun Kwak, Policy-Aligned Predictive Verification for Adaptive VLA Action-Chunk Execution, Under review, 2026.

Seungyeon Kim, Junhoo Lee, Baekseung Kim, Minkyu Kim, Nojun Kwak, Completion Aware Guidance for World Action Models, NeurIPS Workshops – Robotics World Modeling, 2026 (Accepted).

Seungyeon Kim, Junhoo Lee, Minkyu Kim, Baekseung Kim, Nojun Kwak, Recursive Harness Distillation across Agents for Robot Manipulation, arXiv preprint, 2026.

Hyeonjin Kim, Minkyu Kim, Jae Won Jang, KiYoon Yoo, Nojun Kwak, TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Dermatological Diagnosis, Proceedings of the 6th Clinical Natural Language Processing Workshop, 2024.

Minkyu Kim, Suyoung Kim, Hyeonjin Kim, Sunghyun Wee, Junhoo Lee, Baekseung Kim, Nojun Kwak, A Marine Safety Management Model Using Drones, 36th Workshop on Image Processing and Image Understanding(IPIU), 2024.

Ho Seon Choi, Minkyu Kim, Geunwoo Shin, Junki Kim, Inyoung Kim, Yoon Su Baek, Development of the Algorithm for Controlling the Electric Walking Aids on the Slope by Using FSR Foot Sensor, Proceedings of the Korean Society for Precision Engineering Fall Conference, 2017.

Awards and Honors

Cum laude, Seoul National University — Aug 2024
Outstanding BS Thesis Presentation Award, Dept. of Mechanical Engineering, Seoul National University — June 2024
Mechanical Engineering Alumni Scholarship, Dept. of Mechanical Engineering, Seoul National University — Oct 2020
Bronze prize at Mechatronics Competition, Hyundai Infracore & Seoul National University — June 2023
Intel International Science and Engineering Fair 2017, South Korea National Delegation — May 2017

Military Service

The Republic of Korea's Army(ROKA) — South Korea
The 50th Infantry Division — Jan 2021 - July 2022

- Served as Sergeant, Squad Leader and Peer Counselor
- Performed duty as Operational Soldier and Duty Officer

Other Services

Captain of the Engineering College Basketball Team, Seoul National University — 2024
Mentor of "KOTER: Catch-up Orientation Program", Seoul National University — 2022
Vice-President of the Mechanical Engineering Student Council, Seoul National University — 2020
Volunteer Member of "Visiting Engineering Volunteer Group," Seoul National University — 2019
Welfare Division Member of the College of Engineering Student Council, Seoul National University — 2019